

Worksheet 3C: Quadratics — Real-World Applications — Solutions

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: Projectile Motion

1. (a) 15 m (b) 20 m (c) 0 m — back on ground
2. (a) $5t(4 - t)$ (b) $t = 0$ (start) and $t = 4$ (lands)
3. (a) 40 m (b) 25 m (c) $t = 3$ seconds
4. (a) Thrown downward (b) $t = 4$ seconds (c) Speeding up
5. (c) $t \approx 10.4$ s (d) Negative solution is before launch
6. (a) $a = -5$, $b = 20$ (b) $t = 2$ s (c) 20 m
7. (a) $t = 5$ s (b) 145 m
8. (a) 1 m (b) $t = 1.5$ s (c) 12.25 m (d) $t \approx 3.06$ s
9. (a) $y = 15t - 5t^2$ (b) $t = 3$ s (c) 30 m
10. $y = 1.5x - 0.05x^2$
11. $10 = 2 + 12t - 5t^2$; $t = 0.8$ s or $t = 2$ s
12. $t = 1.25$ s

Part B: Area Problems

13. (c) Width 7 m, Length 10 m
14. (c) $w = 25$ m (d) 1250 m^2
15. (b) $s = 5$ cm
16. 13 and 14
17. 16 and 18
18. (c) Width 7 cm, Length 11 cm
19. (c) $x = 3$ cm
20. 8 or -9

Part C: Trig Approximations

- 21. (a) 0.09983... (b) Very close!
- 22. (a) 0.98 (b) Close!
- 23. $\sin(\theta) \approx \theta$ for small θ
- 24. (a) Vertex at $(0, 1)$ (b) Yes, $\cos(0) = 1$
- 25. (a) $\theta < 0.1$ rad (b) $\theta < 0.14$ rad
- 26. (b) 0.1003...
- 27. ≈ 0.9999 (very close to 1)
- 28. (a) 1.105 (b) Very close
- 29. (a) 3 (b) 1.6458... (c) Good!
- 30. Yes, they match

Part D: Probability

- 31. $H^3 + 3H^2T + 3HT^2 + T^3$
- 32. (a) 1 (b) 3 (c) 3 (d) 1 (e) 8
- 33. Each: 0.125, 0.375, 0.375, 0.125
- 34. (a) 0.064 (b) 0.288 (c) 0.432 (d) 0.216
- 35. (a) $6/16 = 0.375$ (b) $5/16 = 0.3125$
- 36. $p^4 + 4p^3q + 6p^2q^2 + 4pq^3 + q^4$
- 37. $10 \cdot (0.25)^3 \cdot (0.75)^2 \approx 0.088$
- 38. ≈ 0.016
- 39. The $6H^2T^2$ term (2 heads) — most likely outcome
- 40. (a) 0.343 (b) 0.784

Part E: Compound Growth

- 41. (a) 1.1025 (b) €1102.50 (c) €102.50
- 42. (b) €1157.63
- 43. (a) 12 years (b) $(1.06)^{12} \approx 2.01$
- 44. $1 + 2r + r^2 = 1.21$
- 45. (a) 1.10 (b) 1.104 (c) When r is small and n is small
- 46. (a) €94.09 (b) €85.87 (c) 23 years
- 47. (a) 53,060 (b) 9.5 years

48. (a) €17,000 (b) €12,282.50 (c) 4.3 years
49. (a) 2 (b) 2.25 (c) 2.594 (d) 2.705
50. (a) 1.127 (b) €468.71 (c) €11,249 (d) €1,249
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